RECOVERY GAIN IS OPERATING-POINT DEPENDENT IN PRUNED TEXT-TO-AUDIO DIFFUSION

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ABSTRACT

Structured pruning is commonly followed by recovery fine-tuning, yet recovery is usually judged at one inference setting. We evaluate released AudioLDM-M pruned and recovered checkpoints across clip duration and prompt domain, then use targeted follow-ups to test why the gain changes. At 83% pruning, CLAP recovery rises from $+0.085$ at 3.84 s to $+0.244$ at 10.24 s and plateaus at 15.36 s. A checkpoint fine-tuned at 3.84 s still gains more when evaluated at 10.24 s, contradicting the prediction of training-duration specialization. Recovery transfers strongly to held-out Clotho captions but is much smaller on hip-hop captions; dense anchors show that the hip-hop battery is not floor-limited. The same duration pattern survives alternative scorers, event metrics, sampler settings, and both pruning severities. These results motivate paired recovery evaluation across operating points rather than a single post-fine-tuning score.

***Index Terms*—** text-to-audio generation, diffusion models, structured pruning, recovery fine-tuning, evaluation

1. INTRODUCTION

Text-to-audio diffusion models have become capable generators, but inference remains expensive because the denoising network is evaluated repeatedly for every clip. AudioLDM-M, for example, uses a 416 M-parameter U-Net [4]. Structured pruning lowers this cost by removing complete channels or blocks, which reduces both parameter count and computation. The smaller network usually loses quality immediately after pruning, so compression pipelines add a recovery stage in which the pruned model is fine-tuned again. This prune-and-recover pattern is established in image diffusion [1–3] and has recently been applied to text-to-audio generation [5].

Most work asks whether the final compressed checkpoint is competitive. A benchmark is chosen, the recovered checkpoint is evaluated at one generation recipe, and the resulting FAD, KL or CLAP value is compared with the dense model. That is a useful systems question, but it does not measure recovery itself. A high final score can arise because the pruned model was already strong, because fine-tuning repaired a large fraction of the pruning damage, or because the adaptation happens to work especially well under the benchmark conditions.

To isolate the recovery stage, we instead study the paired change from the pruned checkpoint P to its recovered counterpart P+FT. We call this change the *recovery gain*. The central question is whether that gain is stable when the same model is used differently. We describe each evaluation condition as an inference *operating point*. Requested clip duration is one axis, prompt domain is another, and sampler configuration provides a third. Under this view, recovery is not a scalar attached to a checkpoint. It is a conditional effect that may change as the operating point moves.

This question fits a broader shift in text-to-audio evaluation. CLAP similarity remains a standard proxy for prompt alignment [8–10], while newer benchmarks increasingly probe robustness and generalization rather than relying on one aggregate score. TTA-Bench is a recent example [23]. Compression evaluation has not yet adopted the same perspective systematically. In particular, a claim that a pruned model "recovers" is usually supported at the same setting used to validate the recovered checkpoint.

We use the released pruned and recovered AudioLDM-M-Full checkpoints of Singh et al. [5] as a controlled case study. The initial experiments reveal a large duration interaction and a much smaller gain on held-out hip-hop prompts. We then test the obvious explanations rather than treating those patterns as mechanisms. A short-duration fine-tune directly asks whether the duration effect follows the duration used during adaptation. A point beyond the native duration tests whether the gain peaks at 10.24 s. Clotho provides a milder held-out domain in which sound-event content remains closer to AudioCaps while caption style changes [20]. Dense anchors test whether the hip-hop battery can resolve alignment at all.

The resulting picture is more useful than a specialization claim. Recovery gain is strongly operating-point dependent, but the duration effect does not follow the fine-tuning duration. Recovery also transfers well to Clotho while remaining small on hip-hop prompts. Our contribution is therefore twofold. We quantify the magnitude and boundary conditions of recovery variation in a pruned TTA model, and we provide a paired evaluation protocol that makes such variation visible without confusing it with final checkpoint quality. Complete provenance and expanded tables remain available in the companion repository [24].

2. BACKGROUND AND MOTIVATION

2.1. Recovery is part of the compression method

Modern diffusion pruning methods rarely stop after structure is removed. Diff-Pruning combines a diffusion-aware pruning criterion with retraining [1]. BK-SDM removes residual and attention blocks and transfers behavior from the larger model through distillation [2]. TinyFusion goes further by selecting shallow architectures according to how well they are expected to perform after fine-tuning [3]. In these systems, the useful compressed model is defined jointly by what pruning removes and what later adaptation can restore.

Singh et al. [5] bring this pattern to AudioLDM by pruning U-Net channels with an L1 criterion and then fine-tuning on AudioCaps for 10^6 steps. Their recovered checkpoints substantially improve the reported in-domain metrics. This establishes that aggressive structural compression can be followed by strong benchmark recovery. It does not establish that the amount recovered is invariant to how the checkpoint is queried.

That distinction matters because fine-tuning is itself an adaptation procedure. Its effect may depend on data distribution, output length, optimization history, or other properties of the pretrained model. Fine-tuning can also improve in-distribution performance while altering behavior under distribution shift [18]. A recovered checkpoint therefore deserves two evaluations. One asks whether it is a good final model. The other asks where the incremental gain from recovery persists.

2.2. Measuring a gain rather than a final score

Automatic TTA metrics expose different aspects of generated audio. CLAP measures text-audio alignment [8]. FAD measures distributional distance but depends on sample size and embedding choice [11–13]. Human-CLAP [15] and event classifiers such as PANNs [14] provide complementary evidence, while FineLAP offers frame-level grounding [16]. None of these metrics turns $S(P{+}FT)$ alone into a measure of recovery.

The useful separation is between three quantities. The final score $S(P{+}FT)$ describes what a user receives. The paired difference $S(P{+}FT) - S(P)$ describes what fine-tuning added. The fraction of a reference gap closed by that difference describes how substantial the addition is relative to the dense model or real audio. These quantities can move differently, so we keep them separate throughout the paper.

3. EXPERIMENTAL DESIGN

3.1. Systems and primary operating points

The dense reference is AudioLDM-M-Full [4], with 415.96 M U-Net parameters. We study two released pruning severities [5]. Severity 1 retains 145.67 M parameters, corresponding to 65.0% pruning. Severity 2 retains 71.08 M, corresponding to 82.9% pruning. At each severity, P denotes the checkpoint before recovery and P+FT the released checkpoint after 10^6 AudioCaps fine-tuning steps. Their paired difference isolates the gain of that recovery stage within the pruned architecture. It does not isolate an effect caused by pruning relative to a dense model given the same fine-tune.

The primary duration comparison uses 3.84 and 10.24 s. Severity 2 adds 5.12 and 7.68 s, yielding a four-point sweep. The original in-domain battery uses AudioCaps TEST [19]. The extreme held-out battery uses hip-hop and rap captions from MusicCaps [21]. Primary severity 2 inference uses 192 AudioCaps prompts and the original hip-hop battery uses 64 prompts. Severity 1 originally uses 80 prompts for duration.

Generation uses DDIM [22] with 50 steps, guidance 2.5 and $\eta = 0$. A pre-specified robustness check repeats the severity 2 endpoint comparison with the published DDIM 200, guidance 3.5 recipe.

3.2. Mechanism and boundary checks

The follow-up experiments were designed after the original analyses and frozen before their own generation and scoring. They are reported as mechanism and boundary-condition checks rather than as part of the original confirmatory family.

Three checks address duration. First, we fine-tune the severity 2 pruned checkpoint for 20,000 steps using 3.84 s random AudioCaps crops and then evaluate that checkpoint at both 3.84 and 10.24 s. If the duration interaction reflects specialization to the training duration, this intervention should reverse its direction. Second, a public dense AudioLDM-M text-fine-tuned checkpoint provides a non-matched reference for whether duration-dependent fine-tuning gains occur without pruning. Third, we extend the released recovery sweep to 15.36 s on 96 prompts. We also add 96 new severity 1 prompts, increasing the pooled duration sample to 176.

The domain follow-up adds Clotho [20], with 96 clips selected before scoring. Its captions are closer in length to AudioCaps than the hip-hop captions, while the dataset remains held out from recovery. We also generate dense references for the hip-hop cells and extend that battery from 64 to 127 prompts. These two checks separate a weak recovery effect from a battery that is simply unable to discriminate text-audio alignment.

3.3. Pairing, endpoints and estimands

Within an operating point, P and P+FT receive identical diffusion noise x_T for each prompt. This common-random-number design removes a large source of sampling variance from the paired difference. The prompt is the statistical unit. Replicates are averaged within prompt before inference, and 95% intervals use a prompt-level percentile bootstrap with $B = 10^4$ resamples.

The primary endpoint is fused CLAP cosine from `laion/clap-htsat-fused`, revision `365dea6e` [8]. A shuffled-caption floor pairs each clip with captions from other prompts in the same battery. Matched dense generations provide a model reference, and real AudioCaps or Clotho recordings provide a data reference when available.

For operating point o, recovery gain is

$$R(o) = \mathbb{E}_i \left[S(P{+}FT, i, o) - S(P, i, o) \right]. \tag{1}$$

The duration interaction is $J = R(10.24\,\text{s}) - R(3.84\,\text{s})$. A positive J means that recovery itself is larger at the longer operating point. We also report

$$\rho_{\text{ref}} = \frac{R}{S(\text{ref}) - S(P)}, \tag{2}$$

which expresses the gain as a fraction of the available gap to a reference.

The original severity 2 duration and domain contrasts and the published-sampler check were specified before their corresponding scores were computed. The four-duration sweep was registered after the primary result but before its own scoring. Human-CLAP, KL, PANNs, anchor decomposition and crop analysis are corroborative or diagnostic analyses and are interpreted accordingly.

4. RESULTS

4.1. Duration changes the amount recovered

The primary severity 2 result is large. Recovery increases from $R = +0.085$ [+0.066, +0.105] at 3.84 s to $+0.244$ [+0.215, +0.273] at 10.24 s, giving $J = +0.159$ [+0.131, +0.188]. The four-point sweep in Table 1 shows a gradual increase rather than an endpoint anomaly. The effect also has practical scale. Recovery closes 44% of the gap from P to dense at 3.84 s and 82% at 10.24 s. Relative to real audio, the corresponding fractions are 33% and 63%.

The robustness evidence is now visible in the paper rather than delegated to the repository. Human-CLAP, KL to real references and PANNs top-10 capture all show the same larger recovery at longer durations. The last 7.68 to 10.24 s increment is not resolved by the secondary metrics, suggesting that most of their increase has already accrued by 7.68 s. The published DDIM 200, guidance 3.5 recipe reproduces the endpoint interaction with $J = +0.184$

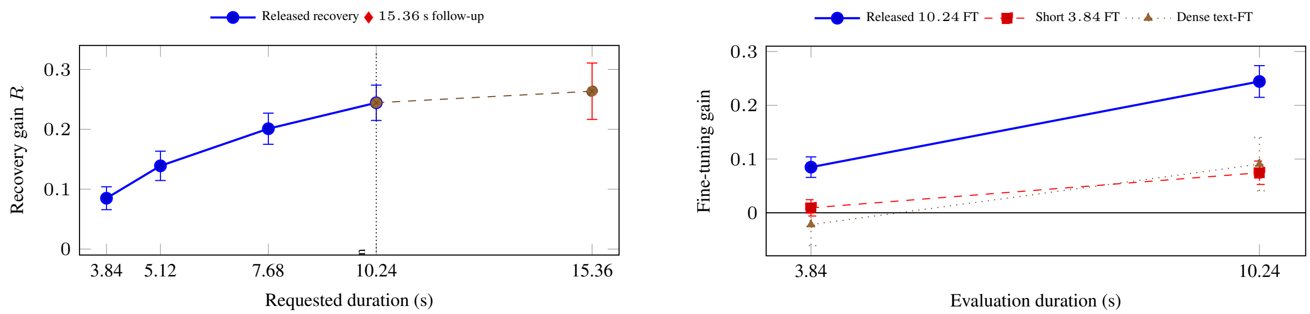

Fig. 1. Duration dependence and its mechanism test. Left, severity 2 recovery gain rises through the original four-point sweep. The 15.36 s follow-up uses 96 prompts and is shown separately because its matched comparison is a subset of the original sweep. The step from matched 10.24 to 15.36 s is unresolved. Right, a checkpoint fine-tuned at 3.84 s still gains more when evaluated at 10.24 s. A public dense text-fine-tuned checkpoint shows the same direction, although it is not a recipe-matched causal control.

Table 1. Recovery across duration and domain. All intervals are 95% prompt-bootstrap intervals. Positive KL recovery means that P+FT reduces KL to the matched real reference relative to P. Human-CLAP, KL and PANNs sweep rescoring are corroborative analyses.

Duration	CLAP R	Human-CLAP R	KL recovery	PANNs capture R
3.84 s	.085 [.066, .105]	.189 [.162, .217]	.66 [.43, .92]	.19 [.06, .32]
5.12 s	.139 [.115, .164]	.278 [.243, .314]	1.16 [.91, 1.41]	.38 [.23, .53]
7.68 s	.201 [.175, .227]	.371 [.336, .405]	1.95 [1.68, 2.23]	.76 [.61, .91]
10.24 s	.244 [.215, .273]	.375 [.340, .409]	2.22 [1.92, 2.52]	.86 [.70, 1.02]
J	.159 [.131, .188]	.185 [.151, .220]	1.56 [1.20, 1.92]	.67 [.49, .85]
Domain	n	$R(3.84\,\text{s})$	$R(10.24\,\text{s})$	$\rho_{\text{dense}}(10.24\,\text{s})$
AudioCaps	192	.085 [.066, .105]	.244 [.215, .273]	.82
Clotho	96	.098 [.072, .125]	.210 [.176, .243]	.74
Hip-hop	127	.026 [.008, .044]	.027 [.004, .051]	.02†

†Hip-hop ρ_{dense} uses the original $n = 64$ anchor subset. On that subset, dense exceeds its shuffled-caption floor by .106 [.079, .135], so the small recovery is not explained by a non-discriminating battery.

[+0.126, +0.243], and a Holm correction over the registered severity 2 family leaves the conclusions unchanged.

Two follow-ups clarify the shape. At 15.36 s, recovery is +0.264 [+0.216, +0.310]. On the matched 96-prompt subset, the step beyond 10.24 s is only +0.021 [-0.023, +0.067]. The gain therefore plateaus rather than peaking sharply at the native duration. Severity 1 also resolves with more data. Adding 96 prompts to the original 80 gives a pooled $J = +0.112$ [+0.076, +0.149]. The new subset has a larger interaction than the original subset, so the pooled value should be read with that between-set heterogeneity in mind.

4.2. The duration effect is not training-duration specialization

The most direct mechanism test changes the duration used for fine-tuning. We fine-tuned severity 2 P for 20,000 steps on 3.84 s crops. At its own training duration, the resulting gain is only +0.009 [-0.006, +0.024]. Evaluated at 10.24 s, the same checkpoint gains +0.075 [+0.053, +0.096], yielding $J = +0.065$ [+0.043, +0.087]. If recovery specialized to the duration on which adaptation was performed, this intervention should have made the short-duration gain larger. It produces the opposite ordering.

A second, weaker reference points in the same direction. A publicly released dense AudioLDM-M text-fine-tuned checkpoint changes by −0.022 [-0.061, +0.017] relative to its dense base at 3.84 s and by +0.091 [+0.042, +0.141] at 10.24 s, with $J = +0.113$ [+0.051, +0.173]. This checkpoint was trained with a different recipe and corpus, so it is not the missing matched dense control. It does show that a duration-dependent fine-tuning gain is not unique to post-pruning recovery.

4.3. Domain transfer is selective rather than absent

The original hip-hop result looked like a failure of transfer, but the expanded domain study changes that interpretation. Recovery transfers strongly to Clotho. At 10.24 s, Clotho gives $R = +0.210$ [+0.176, +0.243] and closes 74% of the dense gap. On a matched 96-prompt comparison, the difference between AudioCaps and Clotho recovery is only +0.032 [-0.023, +0.088]. A shift away from AudioCaps therefore does not by itself erase recovery.

Hip-hop remains different. After extending the battery to 127 prompts, the gain becomes small but resolved, +0.026 [+0.008, +0.044] at 3.84 s and +0.027 [+0.004, +0.051] at 10.24 s. The dense anchors answer the floor objection directly. On the original 64-prompt native cell, dense lies +0.106 [+0.079, +0.135] above shuffled-caption chance, while recovery closes only about 2% of the dense gap. The battery can discriminate alignment, but the recovered checkpoint captures little of the available improvement. Because hip-hop changes both content and caption style, the correct conclusion is selective transfer, not a general in-domain versus out-of-domain dichotomy.

4.4. The short-duration deficit arises during generation

The duration interaction is not explained by where useful events appear in the clip. FineLAP finds the native-duration grounding gain distributed across time, with no early-versus-late difference, $T = -0.002$ [-0.024, +0.020]. A crop analysis reaches the same conclusion from another direction. Scoring only the first 3.84 s of a clip generated at 10.24 s gives $R_{\text{crop}} = +0.172$ [+0.150, +0.194], which is $+0.087$ [+0.065, +0.110] above recovery from a separately generated 3.84 s clip. The deficit therefore arises mainly when the model is asked to generate at the short operating point, not because CLAP sees a short window.

5. DISCUSSION AND LIMITATIONS

The revised experiments sharpen what this study does and does not establish. The robust result is that the incremental gain from fine-tuning depends on the operating point. For the released severity 2 recovery, changing duration changes both the absolute gain and the fraction of the dense gap that is closed. Changing domain can leave the gain nearly intact, as on Clotho, or reduce it by almost an order of magnitude, as on hip-hop. A single post-recovery score cannot represent those behaviors.

The new mechanism tests also rule out the interpretation suggested by the previous title. The duration effect does not track the duration used for recovery fine-tuning. A checkpoint adapted at 3.84 s still benefits more when evaluated at 10.24 s, and the gain stops increasing clearly beyond 10.24 s rather than peaking there. We therefore make no claim that recovery "recovers where it was trained." The evidence is more consistent with duration being a property of how fine-tuning gains are expressed at inference, although the present experiments do not identify the underlying cause.

This distinction is scientifically useful because the two claims lead to different expectations. A specialization account predicts that moving the training duration should move the favorable evaluation duration with it. The short-duration intervention provides exactly that test and the prediction fails. The plateau at 15.36 s removes a second version of the same story in which 10.24 s is a privileged peak. Finally, the dense text-fine-tuned reference shows that a similar length dependence can appear even without a pruning transition. None of these observations identifies a complete mechanism, but together they tell us which mechanism not to report. The phenomenon to reproduce is a duration-dependent fine-tuning gain, not preferential recovery at the training duration.

The domain follow-up makes a parallel correction. The first hip-hop battery mixed a large content shift with much longer captions, so an unresolved gain could not support a general statement about out-of-domain recovery. Clotho supplies an intermediate test and shows that substantial transfer is possible outside AudioCaps. Dense hip-hop anchors then show that the much smaller gain there is not created by a CLAP floor. A useful replication should therefore vary more than one held-out domain and report where each domain sits relative to an informative dense or real-data reference. The result is a transfer profile rather than a binary in-domain versus out-of-domain label.

The missing matched dense fine-tune remains the main causal limitation. Singh's dense checkpoint after the same 10^6-step AudioCaps recovery no longer exists, so we cannot determine whether the duration dependence is specific to recovery after pruning. The public dense text-fine-tuned model is informative but not recipe-matched. The 3.84 s mechanism intervention uses only 20,000 steps, so it establishes the direction of the interaction rather than matching the magnitude of the released million-step recovery.

Other limits concern scope. The study covers one model family and two pruning severities. Clotho improves the domain analysis, but the domain grid is still small, and hip-hop combines a content shift with much longer caption style. The extended severity 1 and hip-hop batteries were follow-ups and show some between-subset heterogeneity. Human-CLAP, KL and PANNs corroborate duration but do not replace perceptual evaluation. The blinded author listening remains a descriptive sanity check and is not used as evidence for the statistical claims.

The practical recommendation is therefore deliberately modest. When a compression method includes recovery, report P and P+FT under at least a few displaced operating points, pair stochastic generations where possible, and express the gain against interpretable references. This adds little conceptual complexity but prevents a local benchmark result from being mistaken for a property of the checkpoint as a whole.

For replication, the most informative unit is the paired prompt rather than the aggregate benchmark score. A reproduction can first verify R at the published operating point, then change one axis while keeping prompts and random seeds aligned wherever the model permits. At least one displaced point should be far enough away to challenge the recovery claim, and intermediate points are useful when the response may be gradual. A dense or real-data anchor is especially important when a null gain could otherwise be confused with a metric floor. If a mechanism is proposed, the cleanest next step is an intervention on that mechanism. Here, changing the fine-tuning duration was more informative than adding another correlational diagnostic because it generated a directional prediction that the data could falsify.

This protocol is intentionally smaller than a general TTA benchmark. It is a way to qualify a recovery claim using the axes most likely to matter for the model under study. Full numerical tables, protocol hashes and executable provenance are retained in the companion repository [24].

6. CONCLUSION

Recovery fine-tuning gain is not constant across operating points in the AudioLDM-M checkpoints studied here. Duration dependence is large and reproducible, but a direct intervention contradicts training-duration specialization as its explanation. Recovery transfers strongly to Clotho and only weakly to hip-hop, whose dense anchors remain well above chance. These results support multi-operating-point evaluation as a more informative way to report recovery after compression.

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